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SSC-I NATIONAL BOOK FOUNDATION FBISE

CHEMISTRY TEST - CHAPTER 19: CHROMATOGRAPHY

Name:		Roll No.		Section:

Total Time: 2:30 Hour
Time Allowed: 150 Minutes

Total Marks: 150

PAPER OBJECTIVE
SECTION-A (MARKS 150)

NOTE: Section-A is compulsory. All parts of this section are to be answered on these pages and handed over to the center superintendent. Deleting / overwriting is not allowed. Do not use lead pencil.

Q.1 Fill the relevant bubble for each part. All parts carry one mark.

Questions	A	B	C	D	A B C D
1. The word chromatography comes from 'chroma', meaning:	Paper	Colour	Light	Writing	○●○○
2. The Greek word 'graphein' means:	Writing	Paper	Colour	Separation	●○○○
3. Chromatography is a very strong _____ tool in analytical chemistry.	Analytical	Building	Cooking	Painting	●○○○
4. Chromatography is used to analyse and _____ compounds.	Burn	Hide	Freeze	Separate	○○○●
5. Chromatography can be coupled with:	Distillation only	Evaporation only	Filtration only	Mass spectrometry	○○○●
6. Chromatography works on the principle of:	Combustion	Oxidation	Neutralization	Partitioning	○○○●
7. In partitioning, components distribute between two phases: stationary and:	Mobile	Solid	Static	Fixed	●○○○
8. The phase that does not move in chromatography is the:	Solvent front	Mobile phase	Stationary phase	Locating agent	○○●○
9. The phase that moves through the stationary phase is the:	Mobile phase	Baseline	Stationary phase	Origin	●○○○
10. The stationary phase may be a solid or a _____ coated on a solid.	Gas	Liquid	Vacuum	Plasma	○●○○
11. The mobile phase is a liquid or a:	Plasma	Gas	Solid	Metal	○●○○
12. In paper chromatography the stationary phase is:	Water adsorbed on paper	The solvent	Air	Ink	●○○○

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Questions	A	B	C	D	A B C D
13. In paper chromatography the mobile phase is a:	Ink	Solvent or mixture of solvents	Water vapour only	Piece of paper	○●○○
14. Different components in chromatography move at different:	Rates	Colours	Temperatures	Masses	●○○○
15. Separation in chromatography is based on affinity for the stationary phase relative to the _____ phase.	Solid	Fixed	Mobile	Gas	○○●○
16. Paper chromatography is used to separate mixtures of inks or some metal:	Atoms	Metals	Ions	Gases	○○●○
17. The filter paper used in paper chromatography is Whatman filter paper number:	Five	Ten	One	Three	○○●○
18. The paper strip in paper chromatography is about _____ cm wide.	25	0.25	10	2.5	○○○●
19. The baseline is drawn in _____ about 5 cm from one end.	Pencil	Pen	Marker	Ink	●○○○
20. The baseline is drawn with a pencil because pencil ink does not:	Dissolve in the solvent	Fade in light	Smell	Dry	●○○○
21. A tiny drop of the mixture is applied at the _____ of the baseline.	Top	End	Midpoint	Edge	○○●○
22. The strip is dried in air for about:	15 seconds	15 minutes	15 hours	15 days	○●○○
23. The strip is suspended so that the impregnated end dips into the solvent to a depth of about:	1 m	50 cm	5–6 cm	5–6 mm	○○○●
24. The chromatography tank is covered to keep the solvent _____ saturated.	Solid	Vapour	Cold	Liquid only	○●○○
25. The strip is removed when the solvent front is _____ below the upper end.	2–3 m	2–3 cm	0 cm	20–30 cm	○●○○
26. The distance travelled by the solvent from the baseline is the:	Solute distance	Solvent front distance	Rf value	Baseline	○●○○
27. Each ink gives coloured _____ at different intervals.	Dots only	Circles only	Bands	Waves	○○●○
28. Rf stands for:	Reaction factor	Retardation (or retention) factor	Rapid factor	Rate formula	○●○○
29. The Rf value is calculated as:	Solute − solvent	Solute × solvent	Distance travelled by solute ÷ distance travelled by solvent	Distance travelled by solvent ÷ distance travelled by solute	○○●○

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Questions	A	B	C	D	A B C D
30. The Rf value is always:	Greater than 10	Less than or equal to 1	Negative	Greater than 1	○●○○
31. The Rf value has:	Units of cm	Units of g	Units of mm	No units	○○○●
32. If the Rf value is greater, the component has more affinity for the:	Baseline	Stationary phase	Mobile phase	Paper only	○○●○
33. A low Rf value means more affinity for the:	Solvent	Stationary phase	Gas	Mobile phase	○●○○
34. If the solvent front travelled 10 cm and the blue component travelled 8 cm, the Rf of blue is:	1.25	0.8	0.08	8	○●○○
35. If the solvent front travelled 10 cm and a purple component travelled 6 cm, the Rf of purple is:	6	0.06	1.67	0.6	○○○●
36. If the solvent front travelled 10 cm and a red component travelled 2 cm, the Rf of red is:	5	2	0.2	0.02	○○●○
37. In the example, the component with greatest affinity for the mobile phase is:	None	Purple	Red	Blue	○○○●
38. In the example, the component with greatest affinity for the stationary phase is:	Purple	Red	None	Blue	○●○○
39. A substance travelling 3 cm while the solvent travels 12 cm has an Rf of:	0.03	0.4	4	0.25	○○○●
40. A substance travelling 6 cm while the solvent travels 8 cm has an Rf of:	0.75	1.33	0.8	0.6	●○○○
41. A substance travelling 4.5 cm while the solvent travels 9 cm has an Rf of:	0.5	2	0.45	4.5	●○○○
42. A substance with an Rf of 0.5 and solvent travel of 12 cm moved:	12 cm	0.5 cm	6 cm	24 cm	○○●○
43. A substance with an Rf of 0.4 and solvent travel of 10 cm moved:	0.4 cm	25 cm	4 cm	10 cm	○○●○
44. An unknown substance is identified by comparing its Rf value with that of:	Known substances	Paper	Water	Air	●○○○
45. Two substances are likely to be the same if their Rf values are:	Greater than 1	Different	Same	Zero	○○●○
46. A pure substance produces _____ spot on the chromatogram.	Only one	Three	Two	No	●○○○

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Questions	A	B	C	D	A B C D
47. An impure substance can produce _____ spots.	More than one	No	Exactly two	Only one	●○○○
48. A substance that locates colourless components in chromatography is a:	Solvent	Baseline	Stationary phase	Locating agent	○○○●
49. Colourless amino acids are located by spraying:	Ninhydrin	Acid	Alcohol	Water	●○○○
50. Ninhydrin reacts with amino acids to give a _____ chromatogram.	Colourless	Black only	Invisible	Colourful	○○○●
51. TLC stands for:	Time liquid chromatography	Thin layer chromatography	Thick layer chromatography	Total liquid chromatography	○●○○
52. Amino acids can be separated by paper chromatography or:	Distillation	TLC	Evaporation	Filtration	○●○○
53. The starting point of the spot on the paper is called the:	Rf	Mobile phase	Solvent front	Origin	○○○●
54. The final position of the solvent is the:	Solvent front	Rf	Origin	Baseline only	●○○○
55. The distance each compound travelled is measured from the:	Edge of paper	Centre	Origin (baseline)	Solvent front	○○●○
56. The retention factor is calculated as distance travelled by compound divided by total distance travelled by the:	Solvent	Spot	Baseline	Paper	●○○○
57. The chromatogram is the:	Tank	Solvent	Locating agent	Developed paper with separated spots	○○○●
58. In chromatography, the solvent rises up the paper by:	Magnetism	Capillary action	Pressure	Gravity	○●○○
59. Which of these is a mobile phase for paper chromatography?	Pencil	A solvent	Clip	Filter paper	○●○○
60. Which of these is the stationary phase in paper chromatography?	Ink	Solvent	Water held on paper	Air	○○●○
61. Chromatography can separate different coloured _____ in a black ink.	Salts	Metals	Dyes	Gases	○○●○
62. Chromatography is a _____ technique.	Physical separation	Combustion	Chemical synthesis	Nuclear	●○○○
63. Chromatography is based on differences in _____ between the components.	Mass only	Affinity	Smell	Colour only	○●○○

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Questions	A	B	C	D	A B C D
64. A component with higher affinity for the stationary phase moves:	Quickly	Not at all always	Backward	Slowly	○○○●
65. A component with higher affinity for the mobile phase moves:	Slowly	Quickly	Not at all	Backward	○●○○
66. The Rf value of a component never exceeds:	100	1	0.1	10	○●○○
67. If a spot does not move from the baseline, its Rf value is:	10	1	0	0.5	○○●○
68. If a spot moves with the solvent front, its Rf value is:	1	0.5	0	10	●○○○
69. Chromatography is useful in forensic science to identify:	Metals only	Inks and drugs	Rocks only	Water only	○●○○
70. Chromatography is used in the food industry to detect:	Food colours	Gases	Rocks	Metals only	●○○○
71. Chromatography is used in medicine to analyse:	Cars	Rocks only	Plastics	Drugs	○○○●
72. The solvent used in chromatography must not _____ the spots.	Move	Spread	Dissolve the paper	Carry	○○●○
73. The unit before Chromatography in the book is:	Separation Techniques	Qualitative Analysis	Hydrocarbons	Biochemistry	○●○○
74. Chromatography is the _____ and last unit of the book.	Eighteenth	Twentieth	Seventeenth	Nineteenth	○○○●
75. Qualitative analysis, separation and chromatography are parts of:	Polymer chemistry	Organic chemistry	Analytical chemistry	Nuclear chemistry	○○●○
76. An ink mixture of blue, black and other colours is placed in a:	Beaker only	China dish	Burette	Pipette	○●○○
77. The ink drops are mixed _____ before application.	Thoroughly	Never	Randomly	Slightly	●○○○
78. The measuring cylinder used in the paper chromatography setup is of _____ capacity.	1 cm^3	100 cm^3	10 cm^3	1 dm^3	○○○●
79. The paper strip hangs from a glass:	Hook	Clip	Pin	Stand	●○○○
80. The solvent in the cylinder is a mixture that must be allowed to stand for _____ before the strip is hung.	1 second	About 15 minutes	1 week	1 day	○●○○

Questions	A	B	C	D	A B C D
81. The strip hangs in the cylinder until the solvent has risen for about:	An hour	A day	A week	A second	●○○○
82. The Rf value helps to identify the components by comparing with:	Mass	Colour only	Random values	Standard data	○○○●
83. A paper chromatogram can show whether a mixture is:	Light or heavy	Old or new	Hot or cold	Pure or impure	○○○●
84. In chromatography, a colourless component can be located by a locating agent or:	Salt	Sugar	Water	UV light	○○○●
85. The principle of chromatography is similar to the way _____ separate coloured dyes.	Burning paper	Freezing water	Heating metals	Water on a tissue	○○○●
86. The mobile phase in gas chromatography is a:	Solid	Plasma	Gas	Liquid	○○●○
87. The mobile phase in liquid chromatography is a:	Plasma	Gas	Solid	Liquid	○○○●
88. In paper chromatography, water molecules are _____ on the paper material.	Adsorbed	Burnt	Removed	Frozen	●○○○
89. Adsorbed means to _____ on the surface.	Evaporate	Dissolve inside	Stick	Burn	○○●○
90. Chromatography was developed mainly as a _____ technique.	Building	Drawing	Separation and analysis	Cooking	○○●○
91. Which is NOT needed for paper chromatography?	A pencil	Solvent	Filter paper	A Bunsen burner	○○○●
92. Which is needed for paper chromatography?	Burette	Distillation flask	Filter paper	Magnet	○○●○
93. Which solvent is commonly used in paper chromatography of inks?	Concentrated acid	Mercury	Oil only	Water or alcohol	○○○●
94. The separated components appear at _____ levels on the paper.	The same	Zero	Different	Infinite	○○●○
95. If two inks give the same Rf, they are probably the _____ compound.	Same	Unknown	Opposite	Different	●○○○
96. The Rf value is characteristic of a compound for a given _____.	Size	Colour	Solvent and paper	Mass	○○●○
97. The Rf of a compound is lowered if its affinity for the paper is:	Higher	Zero	Lower	Equal	●○○○

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Questions	A	B	C	D	A B C D
98. Retardation means:	Slowing down	Heating	Speeding up	Cooling	●○○○
99. The solvent front is marked with a pencil as soon as the paper is:	Removed	Dried	Cut	Inserted	●○○○
100. Chromatography helps to find the number of components in a:	Pure element	Gas only	Mixture	Pure compound	○○●○
101. The chromatogram of a pure compound shows:	Many spots	Two spots	One spot	No spot	○○●○
102. Chromatography separates mixtures by differences in migration:	Names	Masses	Colours	Rates	○○○●
103. If the solvent front travels 15 cm and a spot travels 9 cm, the Rf value is:	6	0.6	0.9	1.67	○●○○
104. If the solvent front travels 20 cm and a spot travels 5 cm, the Rf value is:	0.5	4	0.05	0.25	○○○●
105. If the solvent front travels 8 cm and a spot travels 2 cm, the Rf value is:	4	0.25	0.4	0.2	○●○○
106. If the solvent front travels 12 cm and a spot travels 9 cm, the Rf value is:	1.33	0.75	0.9	0.6	○●○○
107. If the solvent front travels 10 cm and a spot travels 5 cm, the Rf value is:	5	0.05	2	0.5	○○○●
108. If the solvent front travels 16 cm and a spot travels 4 cm, the Rf value is:	0.25	4	0.4	0.64	●○○○
109. If the solvent front travels 10 cm and the Rf is 0.7, the spot travelled:	14.3 cm	0.7 cm	17 cm	7 cm	○○○●
110. If the solvent front travels 20 cm and the Rf is 0.3, the spot travelled:	20.3 cm	66.7 cm	0.3 cm	6 cm	○○○●
111. If the solvent front travels 15 cm and the Rf is 0.2, the spot travelled:	75 cm	15.2 cm	3 cm	0.2 cm	○○●○
112. If a spot travelled 4 cm and its Rf is 0.5, the solvent front travelled:	0.5 cm	4.5 cm	8 cm	2 cm	○○●○
113. If a spot travelled 3 cm and its Rf is 0.25, the solvent front travelled:	0.75 cm	12 cm	1 cm	3.25 cm	○●○○
114. Which Rf value shows the highest affinity for the mobile phase?	0.5	0.2	0.1	0.9	○○○●
115. Which Rf value shows the highest affinity for the stationary phase?	0.8	0.5	0.9	0.1	○○○●
116. Component A has Rf 0.8 and component B has Rf 0.3. Which travels farther?	B	Both equally	Neither	A	○○○●

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Questions	A	B	C	D	A B C D
117. Component A has Rf 0.8 and component B has Rf 0.3. Which is more attracted to the paper?	B	Both equally	Neither	A	●○○○
118. In a mixture of three dyes with Rf 0.2, 0.5 and 0.9, the number of spots is:	9	3	1	2	○●○○
119. If an ink gives four spots, it contains at least _____ components.	4	2	8	1	●○○○
120. If an ink gives one spot, it is probably:	A mixture of two dyes	A mixture of many dyes	Pure	Impure	○○●○
121. Chromatography was first used to separate plant:	Pigments	Gases	Salts	Metals	●○○○
122. Which of these helps to compare an unknown with known compounds on the same paper?	Drawing them separately	Running them side by side	Burning them	Drying them	○●○○
123. The correct order of steps in paper chromatography starts with:	Removing the paper	Spraying ninhydrin	Drawing baseline and spotting the sample	Measuring the Rf	○○●○
124. The solvent level in the tank must be _____ the baseline.	At the level of	Below	Far above	Above	○●○○
125. If the solvent level is above the baseline, the sample will:	Separate well	Move up	Stay fixed	Dissolve in the solvent	○○○●
126. The Rf values depend on the solvent, the paper and the:	Temperature	Time of day only	Colour of tank	Size of spot only	●○○○
127. The stationary phase in thin layer chromatography is a thin layer of:	Water	Oil	Air	Silica or alumina	○○○●
128. Chromatography is considered a _____ technique because only small samples are needed.	Costly only	Slow only	Sensitive	Rough	○○●○
129. When the spots are colourless, the paper can be viewed under:	Fog	Sunlight only	UV light	Dark room only	○○●○
130. Which of the following does NOT affect the Rf value?	The paper	The temperature	The solvent	The colour of the pencil	○○○●
131. A measured Rf value of 1.2 would indicate:	Pure substance	An error in measurement	A slow component	A fast component	○●○○
132. A drug test using chromatography checks for the Rf values of:	Weights	Known drugs	Colours	Sizes	○●○○
133. Black ink separating into several colours shows it is a:	Mixture of dyes	Pure dye	Single compound	Metal	●○○○

Questions	A	B	C	D	A B C D
134. Which diagram shows the baseline?	A line at the top	The paper edge	A line near the bottom of the paper	The solvent front	○○●○
135. The solvent front mark should be made:	Before dipping	After one week	Immediately after removing the paper	Never	○○●○
136. The distance measured to find the Rf of a spot is from the baseline to the _____ of the spot.	Edge	Centre	Bottom of the paper	Top	○●○○
137. A chromatogram of a sample of ink A and a standard dye shows spots at the same height. This suggests:	Ink A contains no dye	Ink A contains the dye	Ink A is a metal	Ink A is pure water	○●○○
138. The retention factor is also known as the:	Ratio to front	Density	Mass ratio	Volume ratio	●○○○
139. Paper chromatography separates amino acids when they are _____ using ninhydrin.	Distilled	Located	Burnt	Mixed	○●○○
140. The paper used in chromatography is best handled by the:	Baseline	Solvent front	Centre	Edges	○○○●
141. Touching the paper with fingers may transfer _____ and spoil the result.	Oils and dirt	Heat only	Sound	Light	●○○○
142. Mixture components move up the paper at different rates because of differing:	Solubilities and affinities	Names	Charges only	Masses only	●○○○
143. Chromatography techniques include paper, thin layer, column and:	Evaporation	Filtration	Fractional distillation	Gas chromatography	○○○●
144. Which technique uses a glass column packed with solid?	TLC	Filtration	Column chromatography	Paper chromatography	○○●○
145. Which technique uses a gas as the mobile phase?	Gas chromatography	TLC	Column chromatography	Paper chromatography	●○○○
146. Which technique is the simplest and cheapest?	HPLC	Gas chromatography	Paper chromatography	Mass spectrometry	○○●○
147. The Rf value of a pure compound is _____ under the same conditions.	Zero	Constant	Random	Variable	○●○○
148. In chromatography the separated components are called:	Metals	Salts	Gases	Fractions or spots	○○○●
149. Chromatography is a vital tool for chemists in industry, forensic science and:	Fashion	Sports	Medicine	Cooking	○○●○

Questions	A	B	C	D	A B C D
150. A pencil line is used for the baseline in chromatography because pencil marks do not:	Show on paper	Dry quickly	Run with the solvent	Smell	○○●○

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